

Introduction to Image Processing

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Image Processing

2026 - 2027

Assessment & Grading

Here's how you'll be graded:

Participation (**Attendant**)/ Assignment/Quizzes
10%

Project (Combine with AI class)
20%

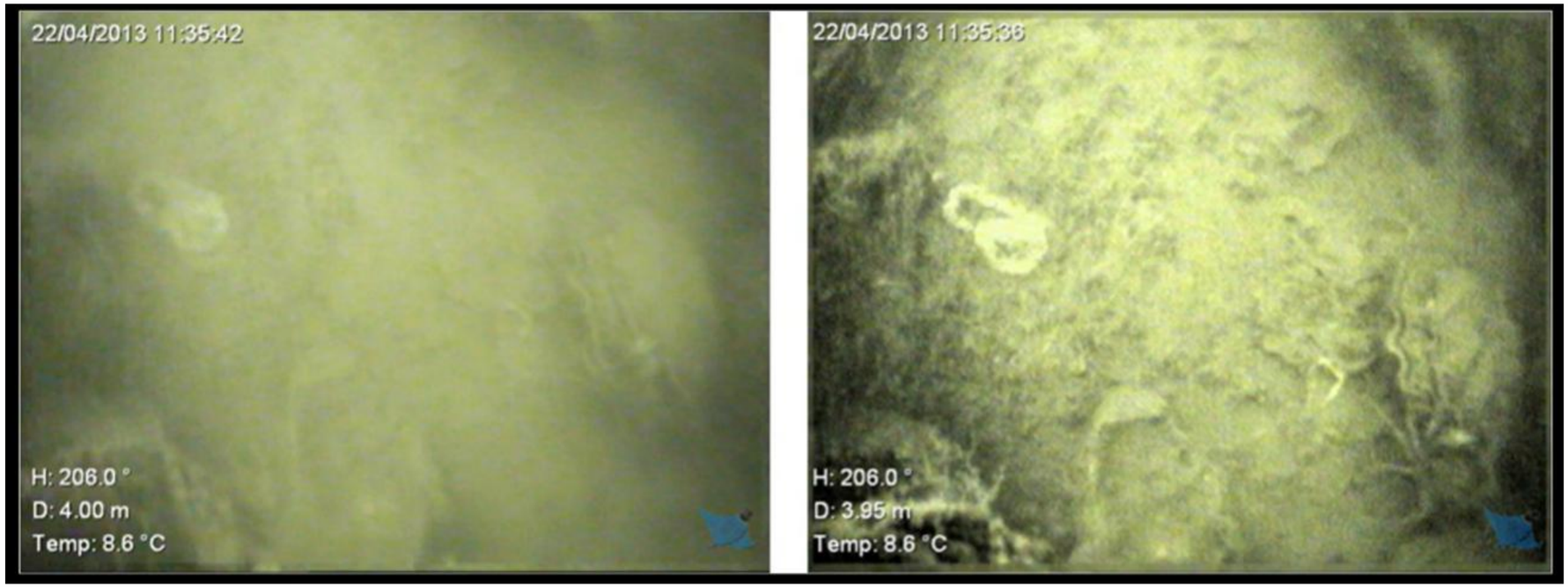
Midterm Exam
20%

Lab (Submit 10% + Clear explain to class 10%)
20%

Final Exam
30%

What is Image Processing?

- Image processing uses computer algorithms to manipulate the image information.



Raw (cannot see anything) Enhance the contrast

Under water image

Image Processing

Enhance quality or extract information from images

Input: Image → **Output:** Enhanced Image

What is computer vision?

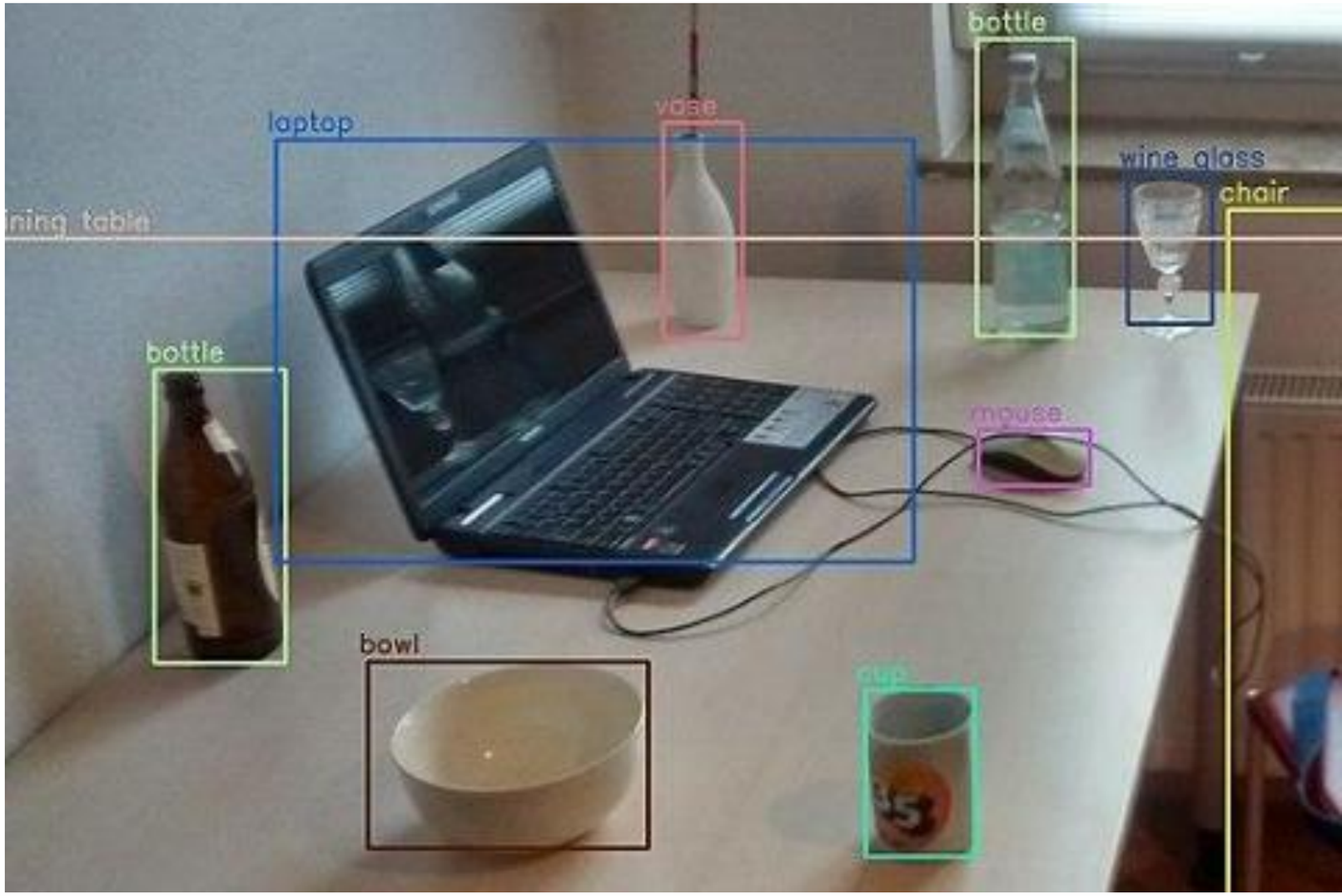
Ability of computers

To understand visual data

For example, images, videos...

Automate tasks

Which human visual system can perform



The Difference Between Processing and Vision

Image Processing





Goal: Quality.
Input is an image,
Output is a better image.

Computer Vision is considered a high-level Image Processing

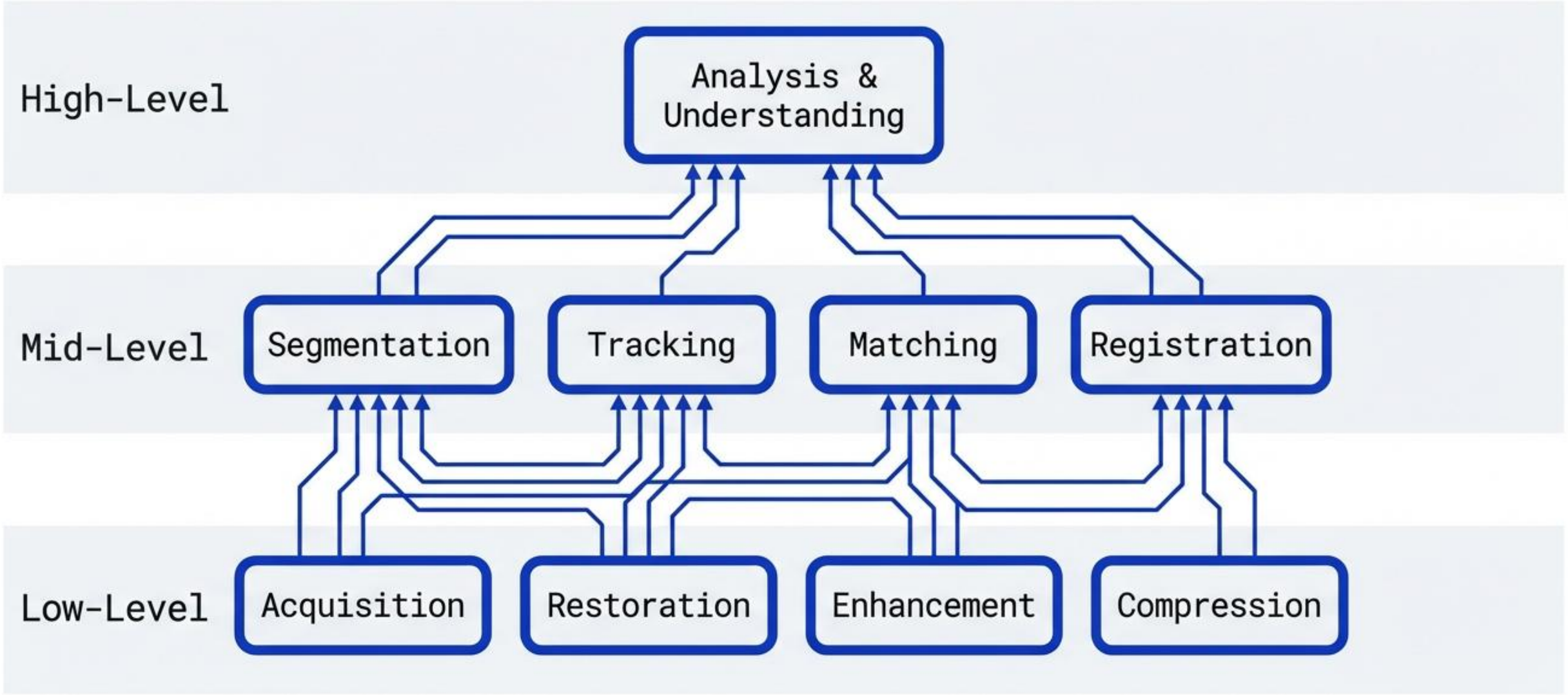
Computer Vision



```
{
  'Object': 'Person',
  'ID': 'Subject_01',
  'Emotion': 'Neutral'
}
```

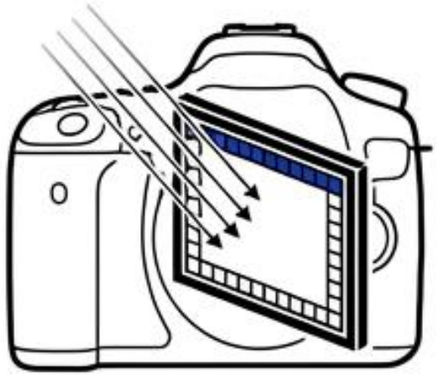
Goal: Insight.
Input is an image,
Output is a symbolic description.

Vision Tasks

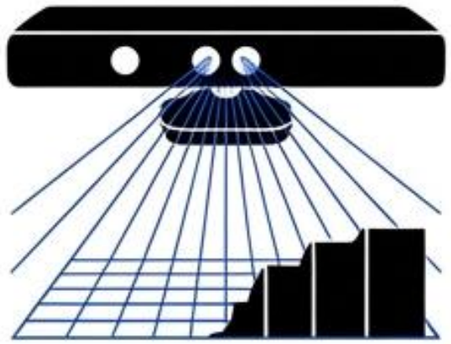


Step 1: Acquisition

Creating digital data from the physical world.



Visible Light / Video



Depth & Range



Volumetric / Medical

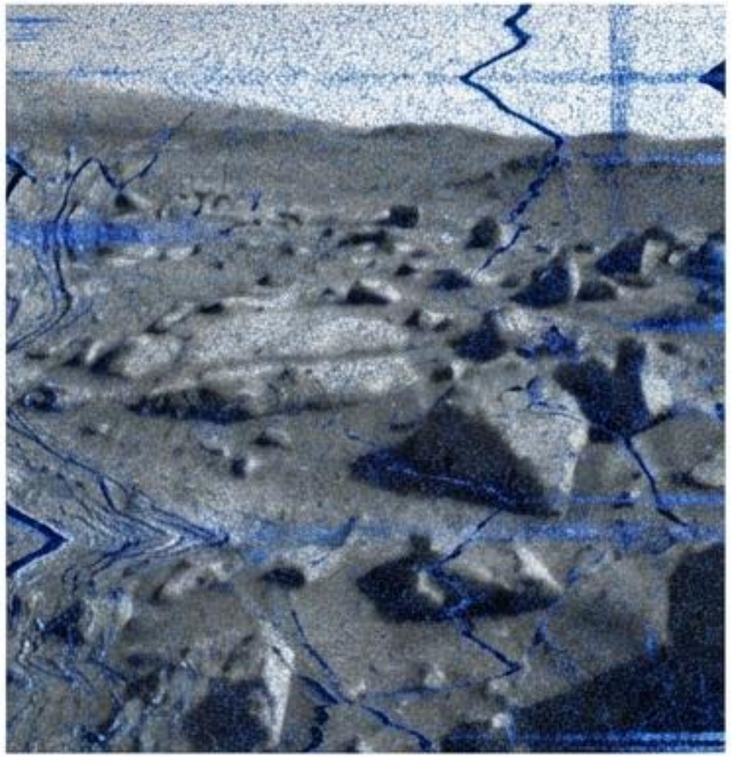


Sound Wave

The conversion of physical signals into high-dimensional digital arrays.

Restoration: Reversing the Damage

Before and After



Degraded Input

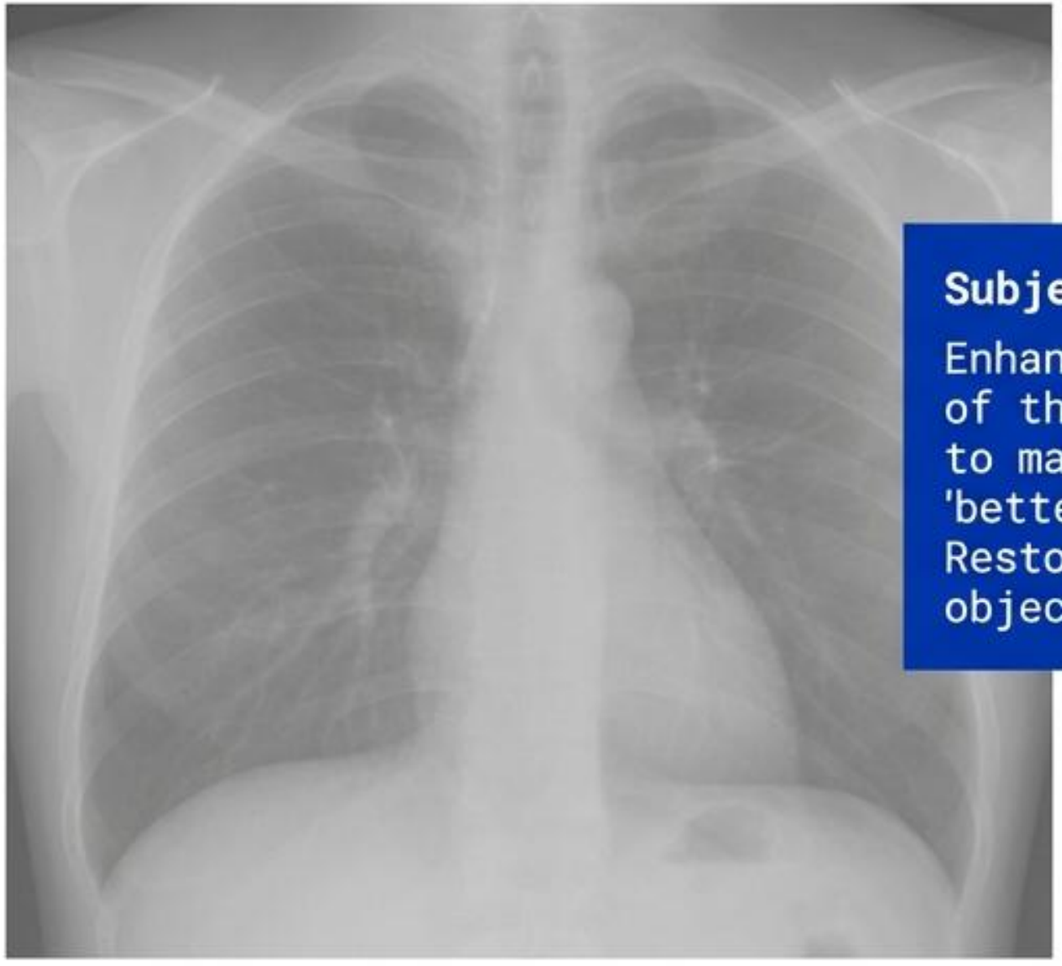
Inverse Model of Distortion



Restored Output

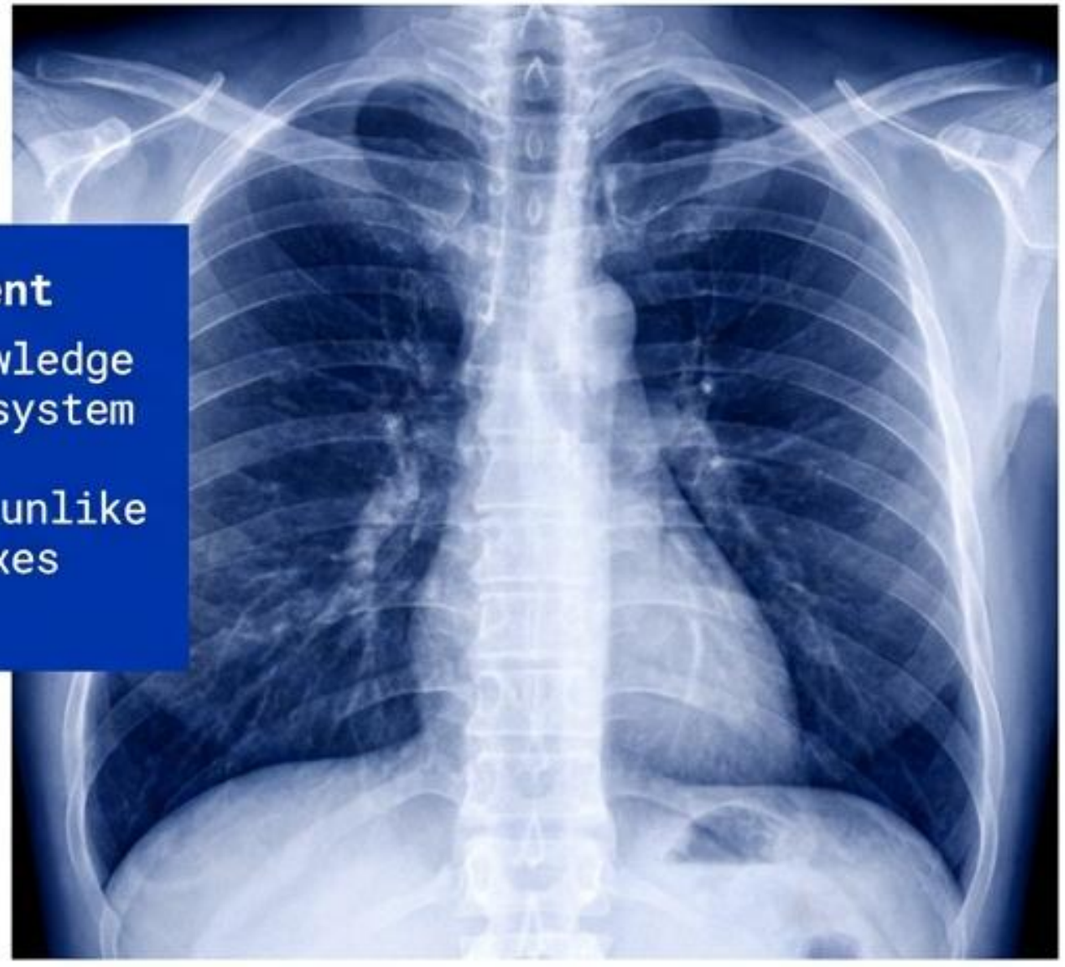
Restoration models the degradation process (blur, noise, distortion) and applies the inverse to recover the original appearance. It is an objective process.

Enhancement: Pleasing the Eye



Subjective Improvement

Enhancement uses knowledge of the human visual system to make images look 'better' to our eyes, unlike Restoration which fixes objective errors.



Raw

Enhanced

Enhancement: Pleasing the Eye



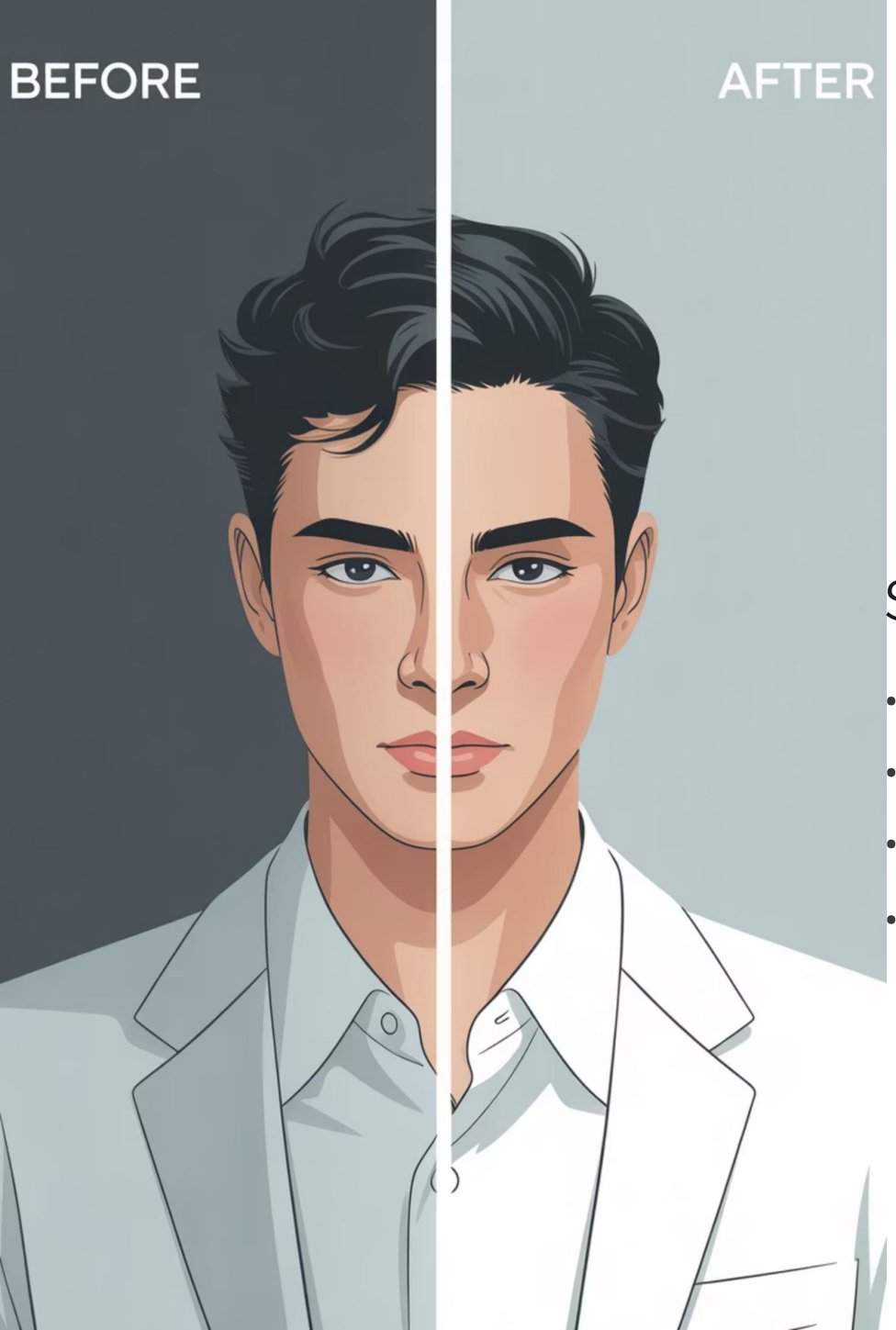


Image Enhancement Techniques

Spatial Domain Methods

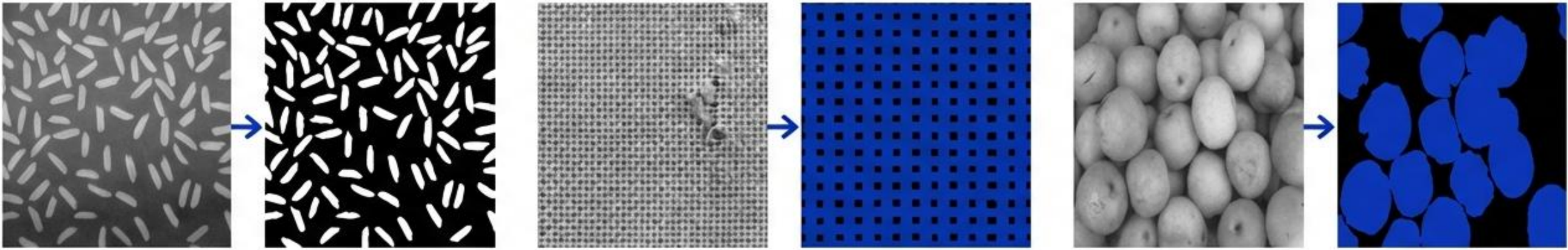
- Direct pixel manipulation
- Histogram equalization
- Contrast stretching
- Brightness adjustments

Frequency Domain Methods

- Fourier transform analysis
- Low-pass filtering (smoothing)
- High-pass filtering (sharpening)
- Band-pass filtering

Segmentation: The Art of Separation

Separating objects from one another and the background.



Visual Perception

> The process of acquiring knowledge about environmental objects and events by extracting information from the light they emit or reflect

Vision → **Understanding** → **Knowledge**

📷 **Perception is like taking a picture.**

Whenever you are watching something, it doesn't mean you are perceiving it.

Visual Perception

What can you see in this picture?



Visual Perception

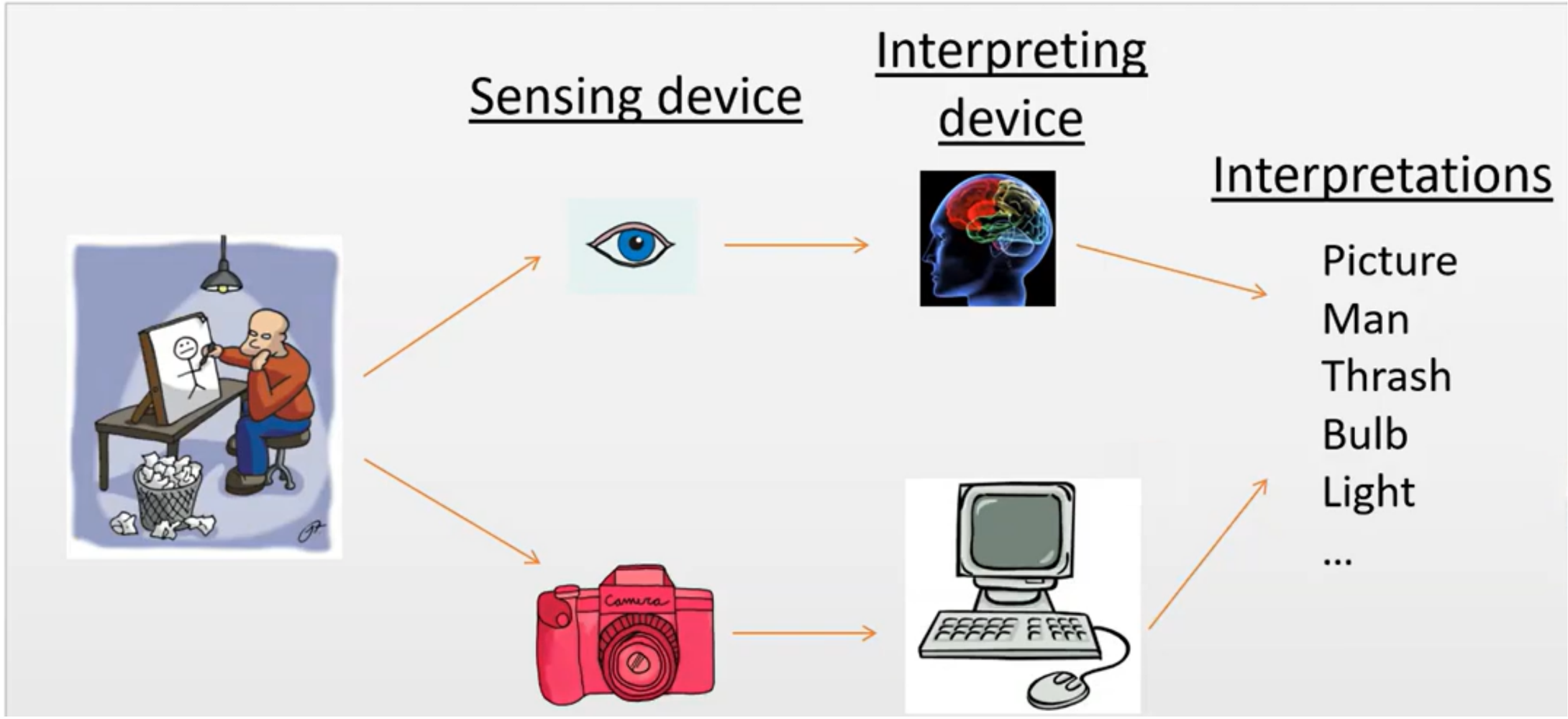
Now, what can you see in this picture?



The process of visual perception

It is not just what you are seeing but whether you are interpreting it correctly or not

Human vision vs. Computer vision



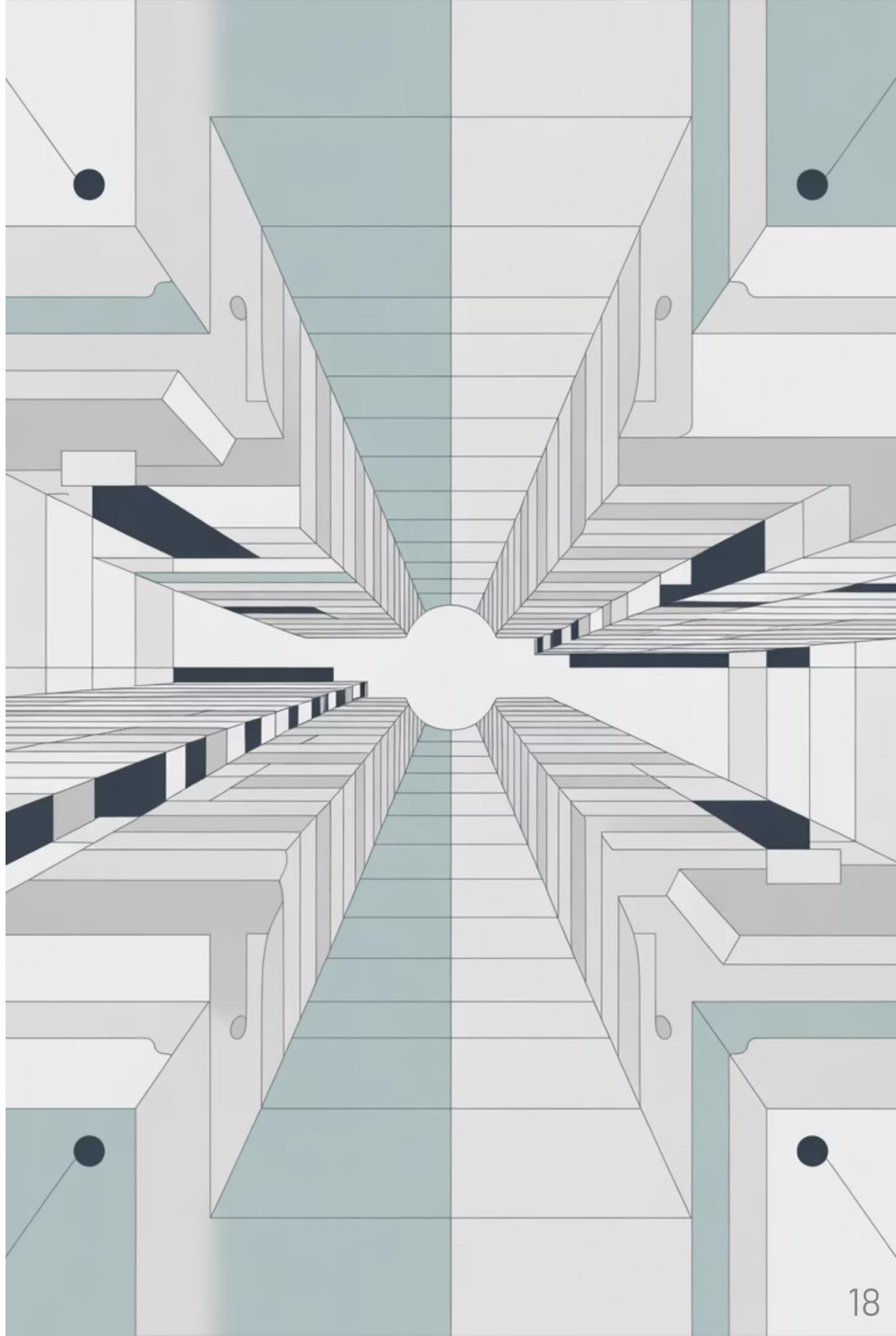
Every image tells a story



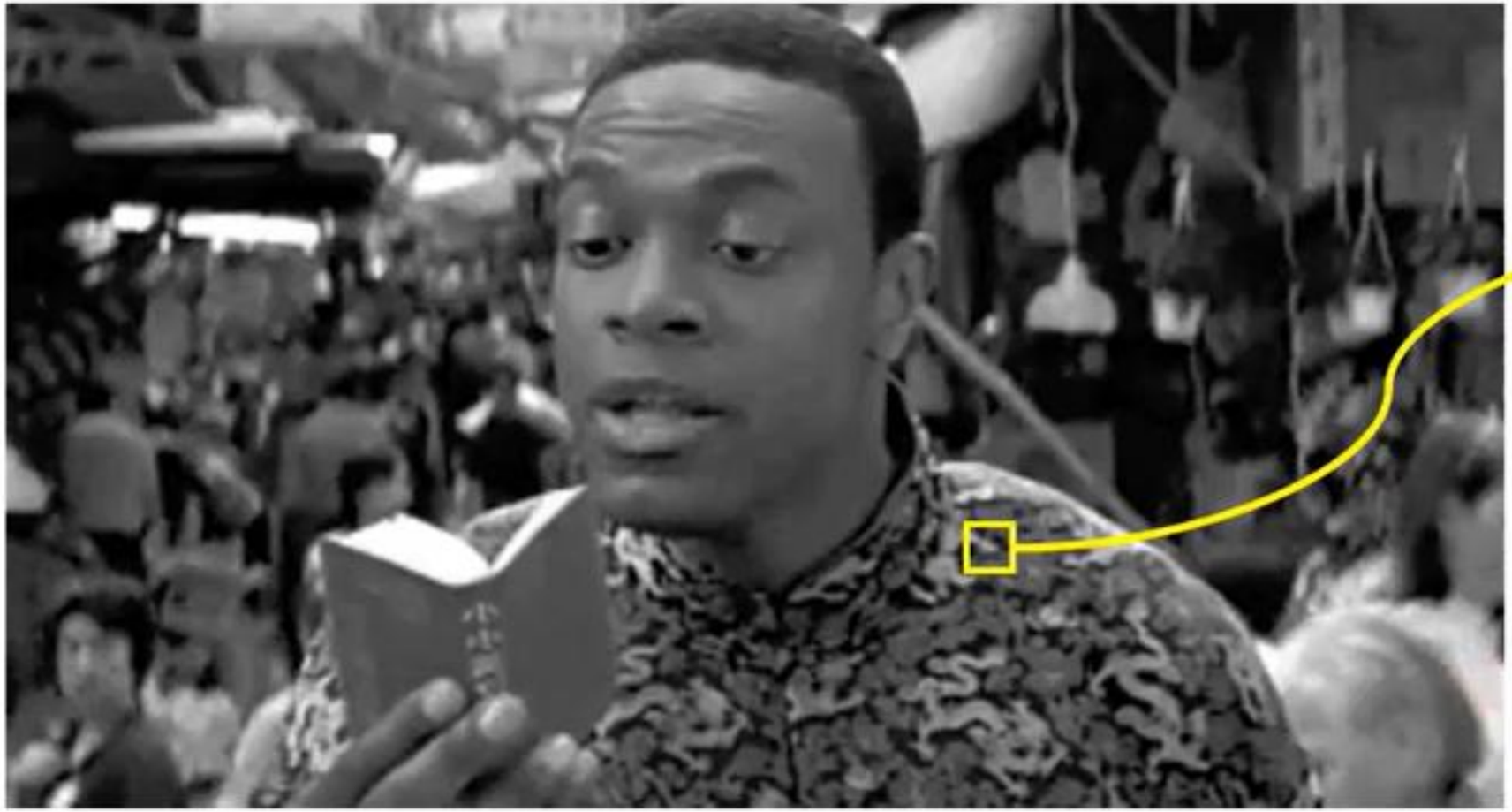
- Goal of computer vision: perceive the “story” behind the picture
- Compute properties of the world
  - Shape
  - Names of people or objects
  - What happened?

How Computers See Images

- Humans see images as smooth, meaningful scenes.
- Computers see images as grids of numbers.
- Each image is stored as a matrix of pixels, where each pixel holds numerical values representing color intensity.

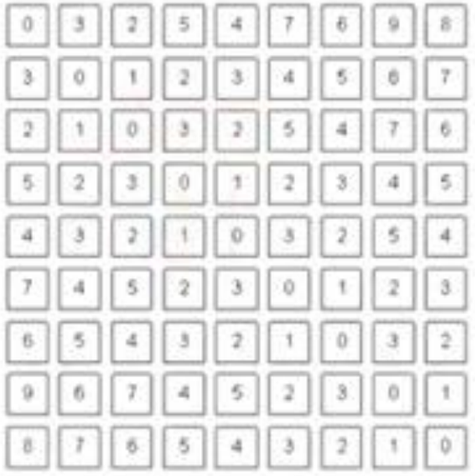


How Computers See Images



We see



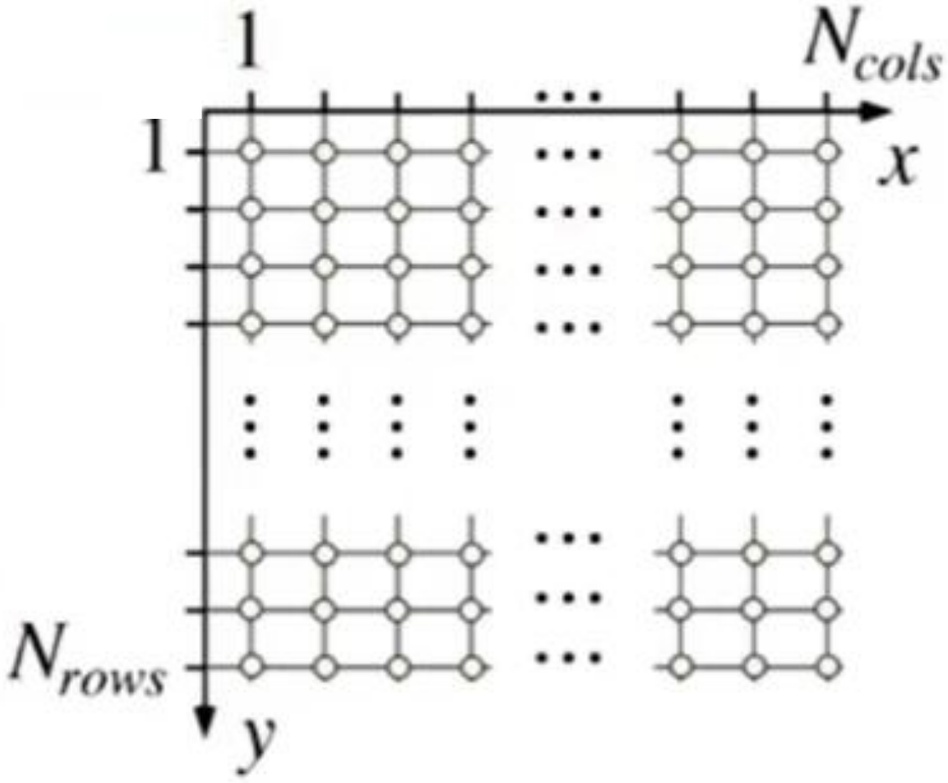


Computer See

What is digital image?

The sampling of continues signal that is presenting in 3D world into a spatial domain





The Digital Image Foundation

Pixels

The fundamental building blocks tiny colored squares that form the complete image

Resolution

Dimensions measured in width × height, determining image detail and clarity

Color Depth

Bits per pixel defining the range of possible colors and tones

Color Depth	Bits per Pixel	Possible Colors
1-bit	1	2 (black & white)
8-bit	8	256 (grayscale)
24-bit RGB	8 bits per channel × 3	16.7 million colors

Resolution Matters

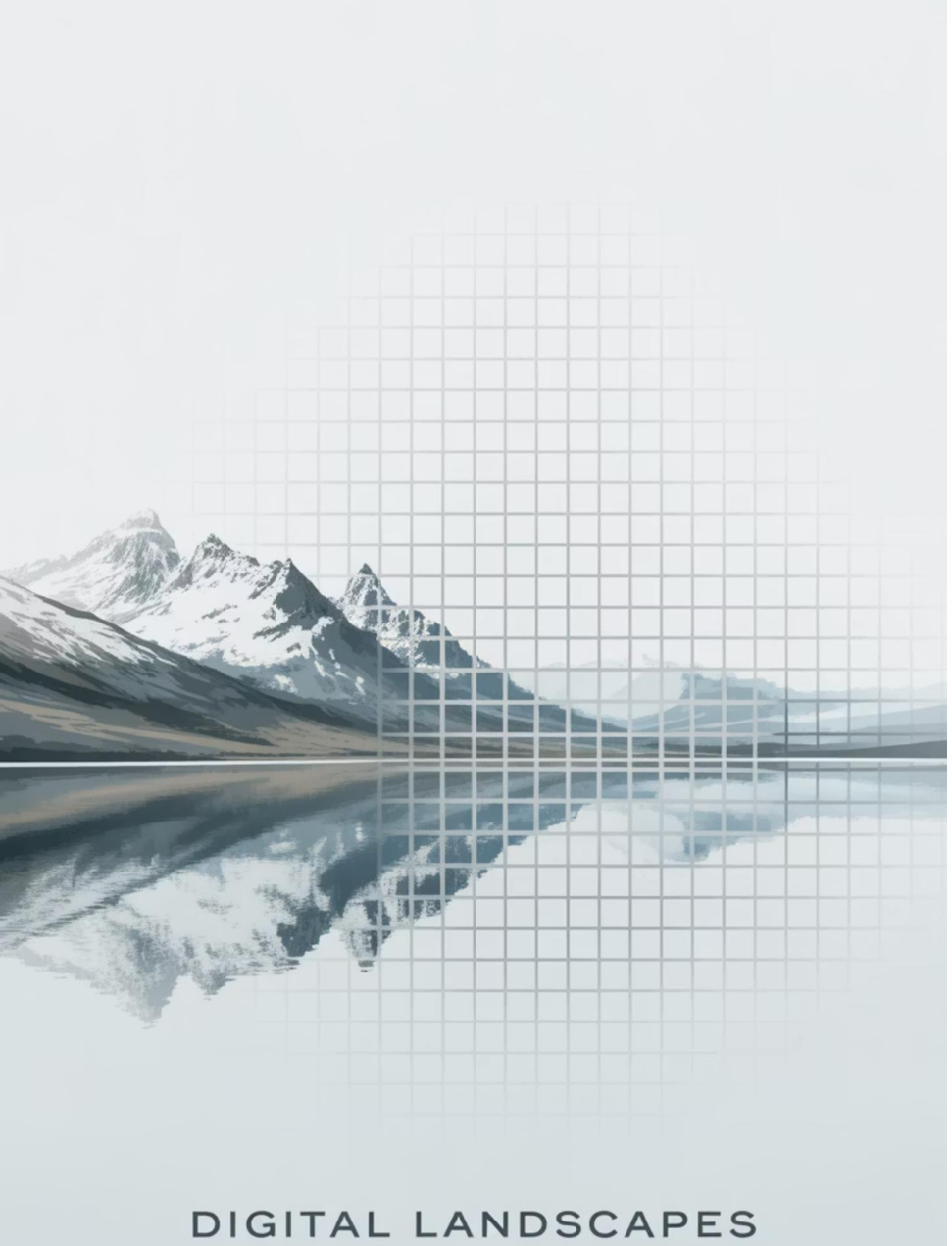
Spatial Resolution

The number of pixels in an image determines the level of detail captured.

Example: 1920×1080 (Full HD) vs. 3840×2160 (4K)

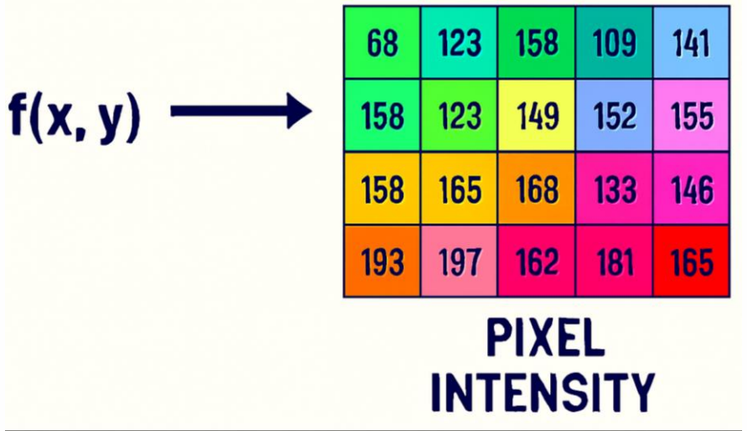
Higher spatial resolution = More detail, but larger file sizes



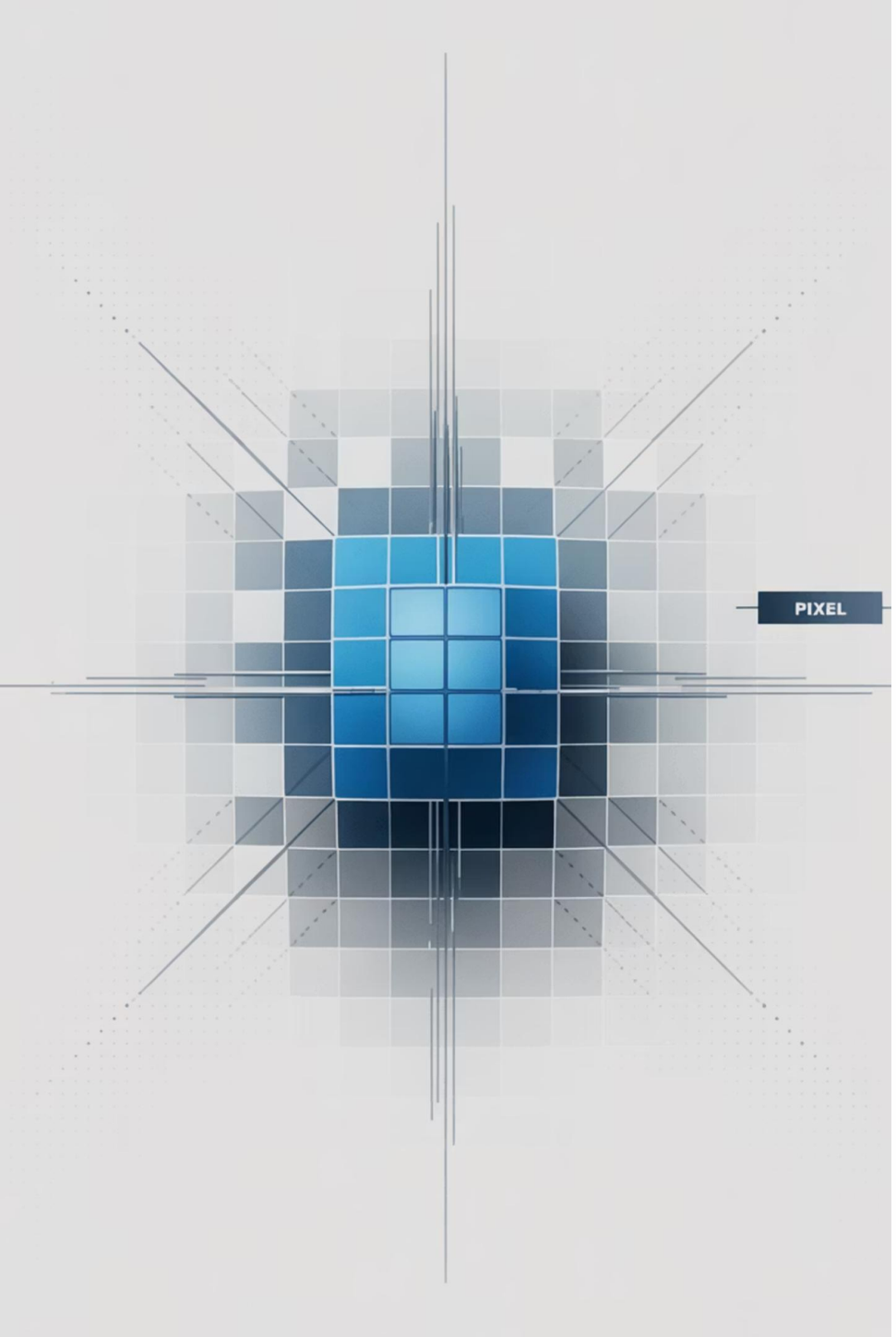


The Digital Image Foundation

- An image can be mathematically described as function $f(x,y)$ or a grid of pixels arranged in rows and columns
- Each coordinate pair (x,y) corresponds to a point in space, and the function's value represents the intensity (brightness) at that location
- Each pixel holds information about color and brightness, allowing the image to visually represent patterns, textures, and objects



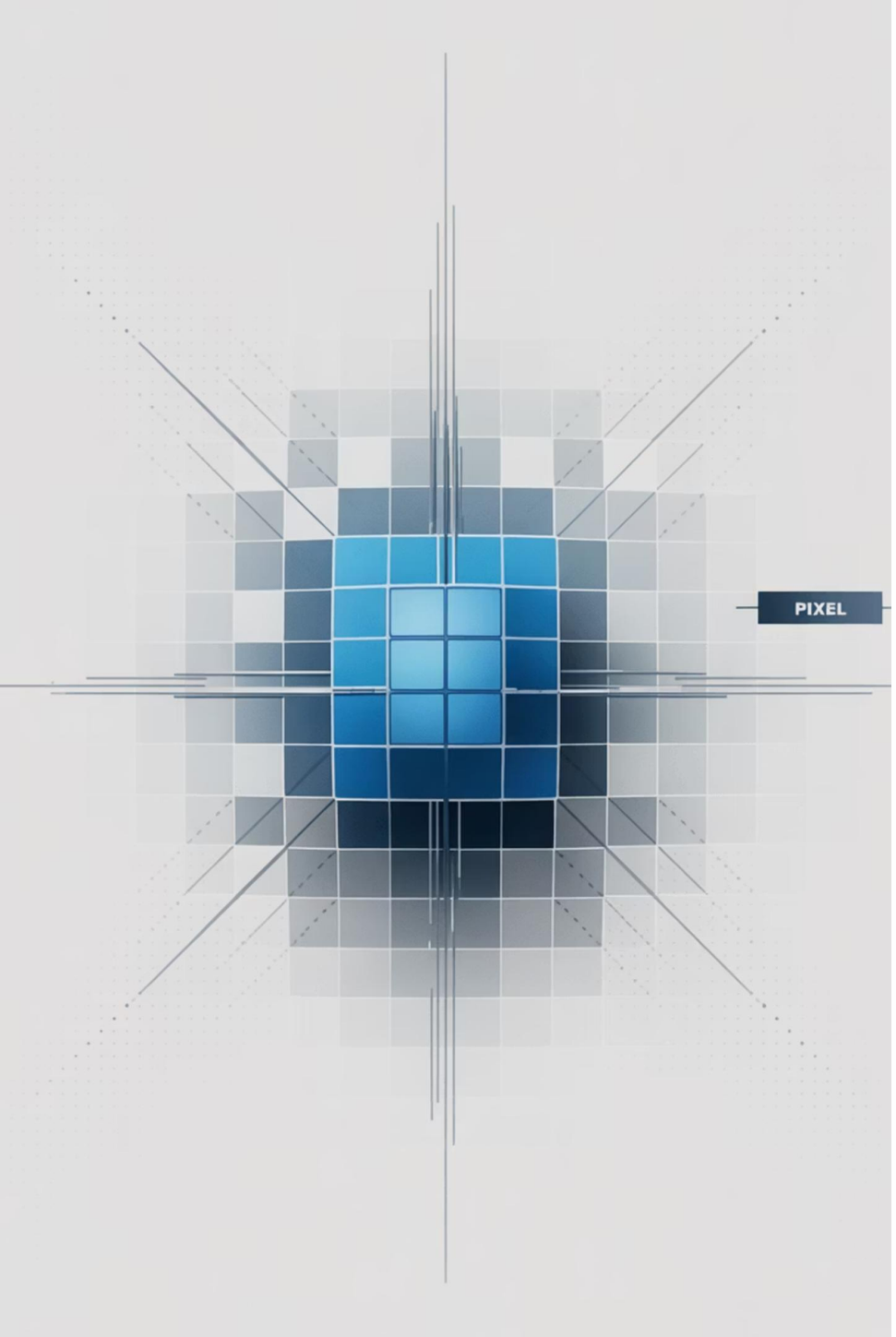
Usually in RGB format (Red, Green, Blue)



Pixel Neighborhoods & Connectivity

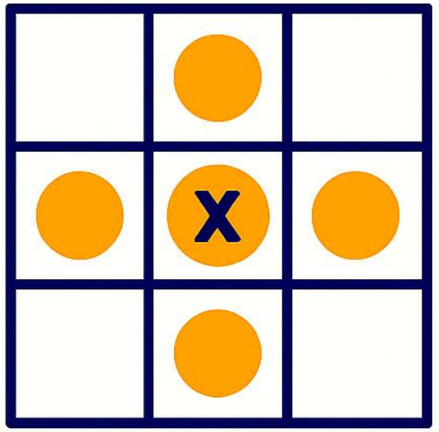
- Image pixels don't exist in isolation. The relationship to their neighbors is crucial for image processing tasks like object detection and filtering.
- **Pixel Neighborhoods**: This refers to the pixels immediately surrounding a central pixel.
  - ✓ 4-Neighbors: The four pixels directly adjacent (**up, down, left, right**).
  - ✓ 8-Neighbors: All eight pixels that touch the central pixel, including the diagonals.
- **Pixel Connectivity**: This concept defines whether connected pixels belong to the same object.
  - ✓ 4-Connectivity: Pixels are considered connected only if they are 4-neighbors.
  - ✓ 8-Connectivity: Pixels are considered connected if they are 8-neighbors, allowing for diagonal connections.

Pixel Neighborhoods & Connectivity



PIXEL NEIGHBORHOODS & CONNECTIVITY

4-CONNECTED



8-CONNECTED

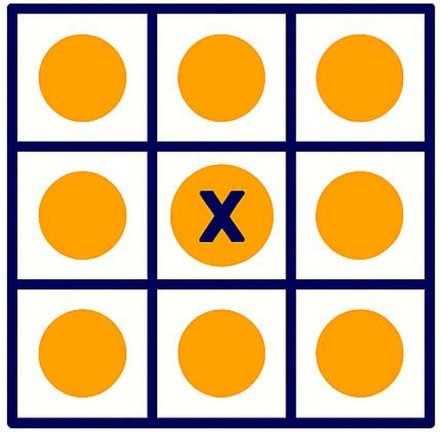
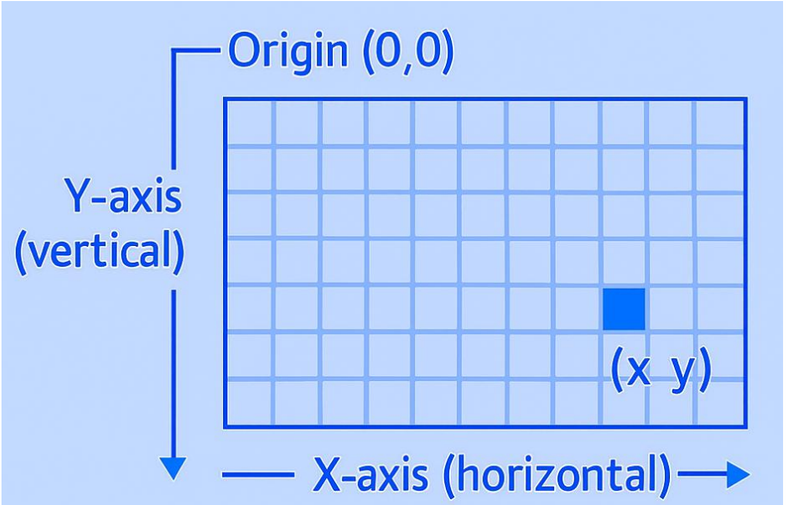
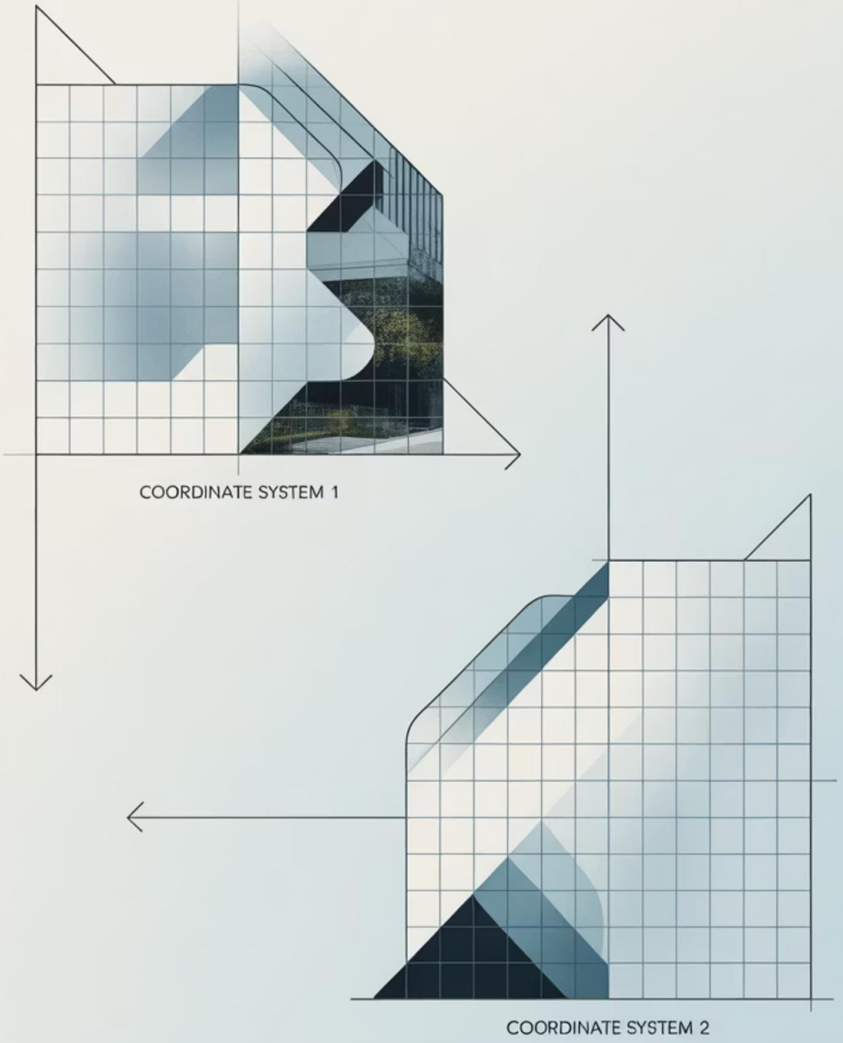


Image Coordinate Systems

To work with pixels, we need a way to refer to their specific location within an image grid.

- **The Origin (0,0):** In digital imaging, the coordinate system almost always starts at the top-left corner of the image.
- **X-axis (Columns):** This axis runs horizontally from left to right. As you move to the right, the x-coordinate increases.
- **Y-axis (Rows):** This axis runs vertically from top to bottom. As you move down, the y-coordinate increases.





Types of Digital Images

Color RGB Images

Three channels (Red, Green, Blue) each with 0-255 values. Most common format for photographs and natural images. Total of 16.7 million possible colors.

Grayscale Images

Single channel with intensity values typically ranging from 0-255. Common in medical imaging, document processing, and as a preprocessing step for color images.

Binary Images

Only two possible values: 0 (black) or 1 (white). making it extremely memory-efficient and ideal for tasks that require clear separation between foreground and background, though it provides limited visual detail.

Color Depth	Bits per Pixel	Possible Colors
1-bit	1	2 (black & white)
8-bit	8	256 (grayscale)
24-bit RGB	8 bits per channel × 3	16.7 million colors

Visual Comparison of Image Types

RGB



Grayscale

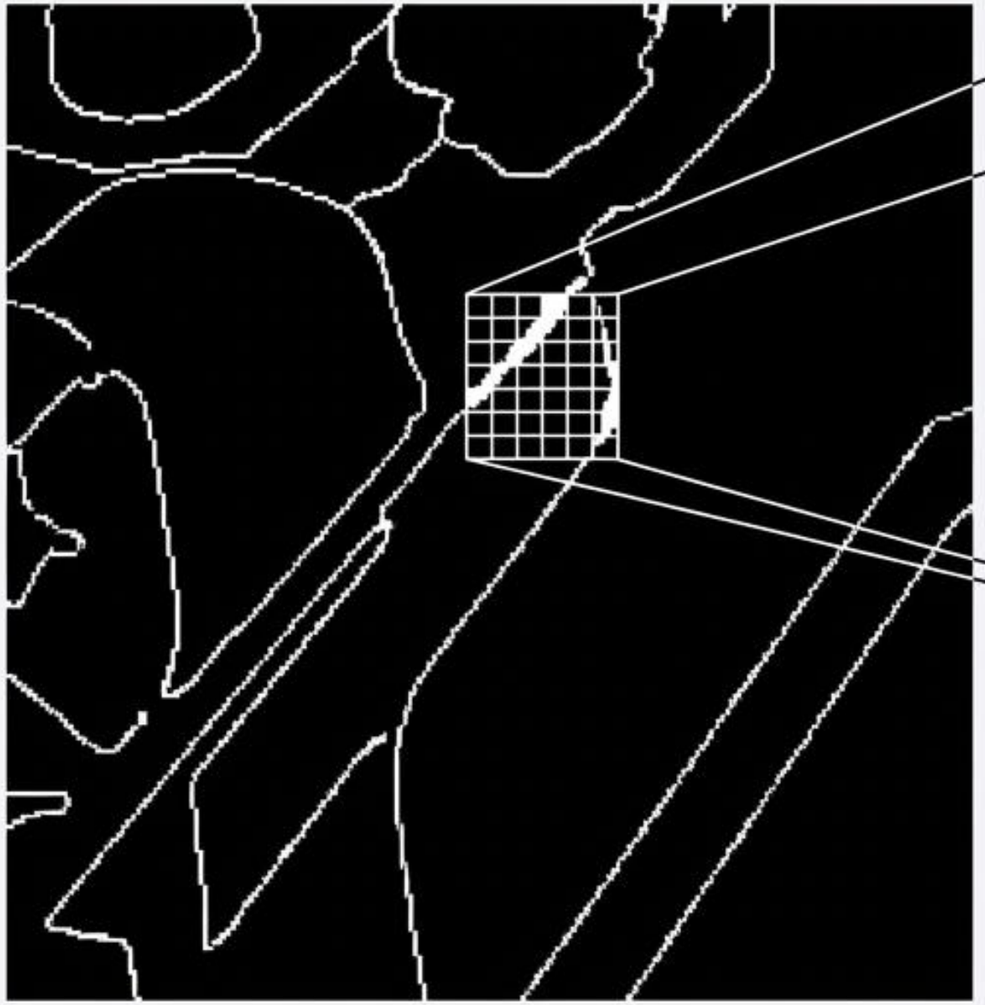


Binary



Binary Images

The Logical Image

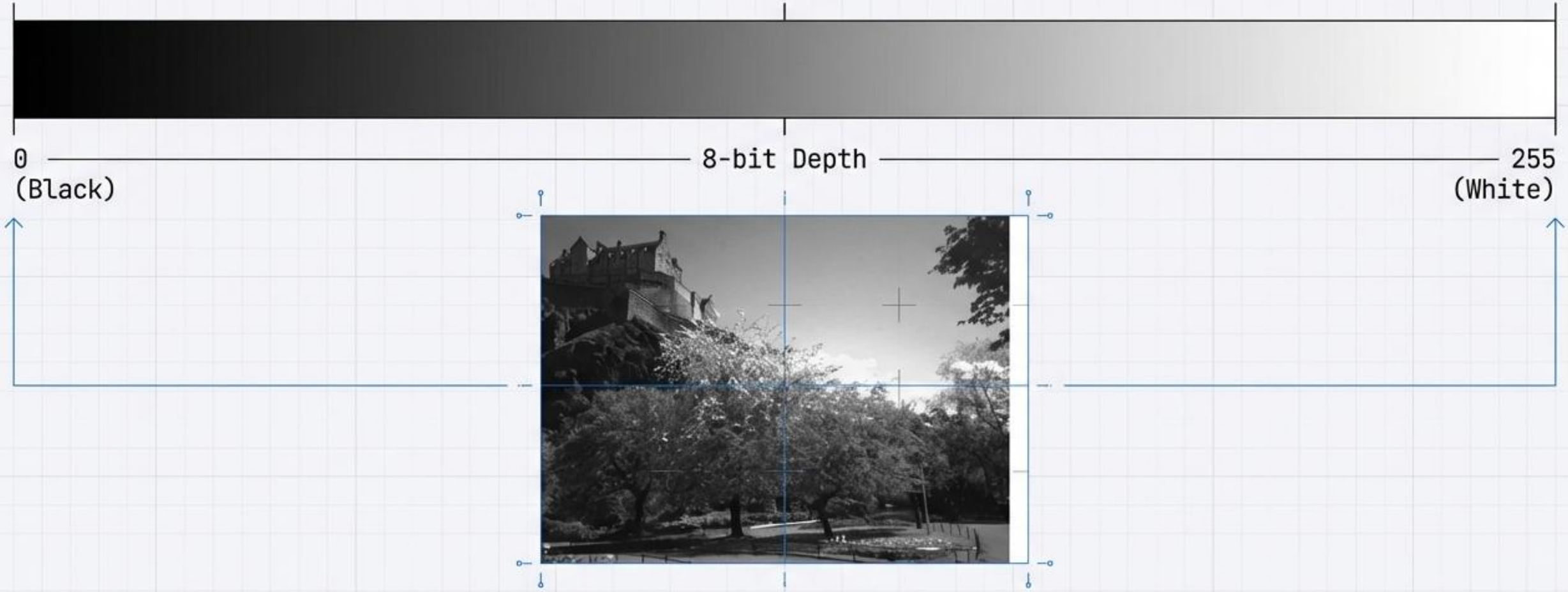


1	1	1	1	1	1	1	1
0	0	0	1	1	1	1	1
0	0	0	0	0	0	1	1
0	0	0	0	0	0	1	1
0	0	0	0	0	0	1	1
0	0	0	0	0	0	1	1
0	0	0	0	0	0	1	1
0	0	0	0	0	0	1	1

Value Set: {0, 1}
0 = Black (Background)
1 = White (Foreground)
Primary Use: Segmentation & Masks

Grayscale Images

Intensity Maps



Single Channel Array. Each pixel represents light intensity.

Color (True-Color) Images

The Layer Cake Deconstruction

The RGB Color Space



49	55	56	57	52	53
58	60	60	58	55	57
58	58	54	53	55	56
83	78	72	69	68	69
88	91	91	84	83	82
69	76	83	78	76	75
61	69	73	78	76	76

Red Channel (R)

64	76	82	79	78	78
93	93	91	91	86	86
88	82	88	90	88	89
125	119	113	108	111	110
137	136	132	128	126	120
105	108	114	114	118	113
96	103	112	108	111	107

Green Channel (G)

66	80	77	80	87	77
81	93	96	99	86	85
83	83	91	94	92	88
135	128	126	112	107	106
141	129	129	117	115	101
95	99	109	108	112	109
84	93	107	101	105	102

Blue Channel (B)



Grayscale

- Grayscale can be considered a simplified color space.
- It represents images using only shades between black and white with the range of (0 - 255) grays level.
- Unlike RGB which use multiple channels, grayscale images are typically represented by a single channel, where each pixel's value indicates its brightness or intensity.
- Grayscale images are often created by converting color images, typically by averaging the R, G, and B values.
- Its simplicity makes grayscale fundamental in various image processing techniques, from edge detection and noise reduction to feature extraction in computer vision.



Binary

- The binary representation, often referred to as a bi-level image, is the simplest form of digital image representation.
- Each pixel in a binary image can only have black or white, representing 0 or 1.
- Due to its simplicity, binary images are incredibly memory-efficient, requiring only one bit per pixel. This makes them fast to process and store.

Color Space Conversion

- **RGB → Grayscale:** Grayscale conversion reduces color complexity, making images easier to process and analyze. It helps highlight important features and prepares images for tasks like edge detection or pattern recognition by focusing only on brightness, not color.

Luminosity Method

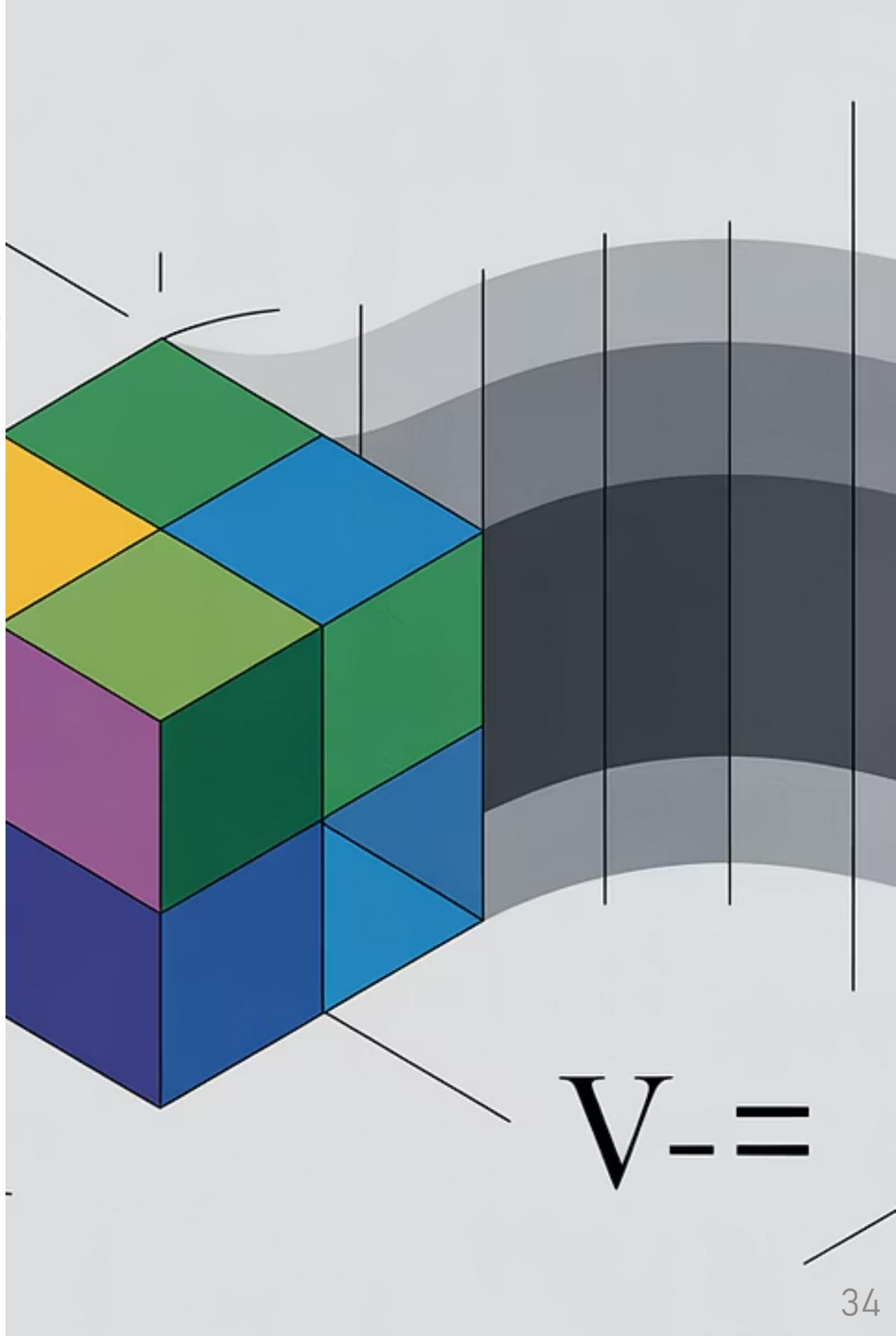
This is the most common and perceptually accurate method, simulating how humans perceive brightness. It uses a weighted average: `Grayscale = 0.299R + 0.587G + 0.114B`

Average Method

The simplest approach, it calculates the arithmetic mean of the R, G, and B components: `Grayscale = (R + G + B) / 3`. While easy to implement, it often produces less accurate perceptual results.

Lightness Method

This method determines grayscale by averaging the maximum and minimum color components: `Grayscale = (max(R, G, B) + min(R, G, B)) / 2`. It provides a balance between simplicity and perceptual accuracy.



Color Space Conversion

Binarization is a fundamental point processing technique that converts a grayscale image into a binary image, containing only two pixel values (typically black and white). This simplification is crucial for isolating objects and simplifying complex scenes for easier analysis.

Convert to Grayscale

The first step is to transform the color (RGB) image into a grayscale image. This reduces each pixel's data from three color channels to a single intensity value.

Apply a Threshold

A threshold value is then chosen. Each pixel in the grayscale image is compared to this threshold. Pixels above the threshold are set to white (255), and those below are set to black (0).

Result: Binary Image

The output is a binary image, where objects of interest are typically represented in white against a black background, or vice-versa, making them easy to identify.

The conversion process involves two main mathematical operations:

1. RGB to Grayscale Conversion:

2. Thresholding for Binarization:

$$B(x,y) = \begin{cases} 255 & \text{if } I(x,y) > T \\ 0 & \text{if } I(x,y) \le T \end{cases}$$

Where B is the binary pixel value, I is the grayscale intensity, and T is the chosen threshold value.